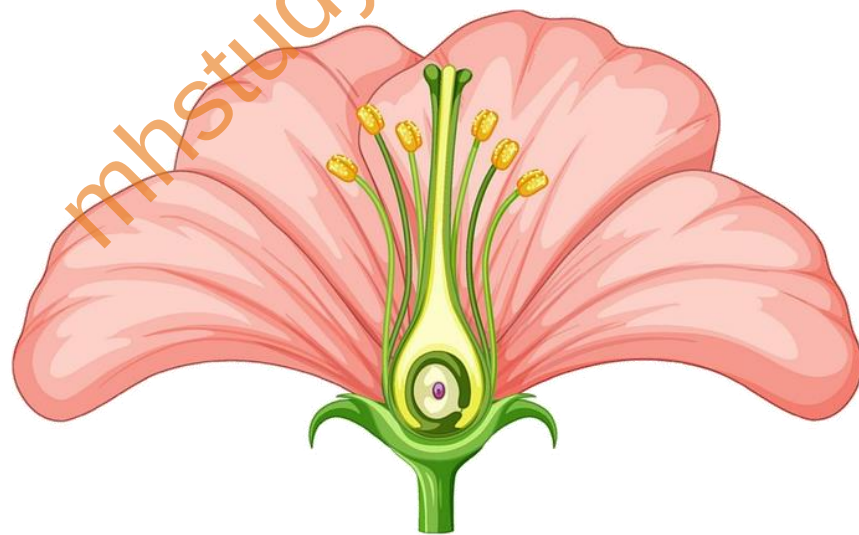


Chapter 3

LIFE

PROCESSES - 2



1. Complete the following chart.

Asexual reproduction	Sexual reproduction
Reproduction that occurs with the help of somatic cells is called as asexual reproduction.	Reproduction that occurs due to formation of gametes is called as sexual reproduction.
Male and female parents are not necessary for asexual reproduction.	Male and female parent are necessary for sexual reproduction.
This reproduction occurs with the help of mitosis only.	This reproduction involves meiosis during gamete formation.
New individual formed by this method is genetically identical to the parent.	New individual formed by this method is genetically different from parents
Asexual reproduction occurs in different individuals by various methods like binary fission, multiple fission, budding, fragmentation, regeneration, vegetative propagation, spore production, etc.	Sexual reproduction occurs by two main processes - gamete formation and fertilization.

2. Fill in the blanks.

- a. In humans, sperm production occurs in the organ testis.
- b. In humans, Y chromosome is responsible for maleness.
- c. In male and female reproductive system of human, no gland is same.
- d. Implantation of embryo occurs in uterus.
- e. Asexual type of reproduction occurs without fusion of gametes.
- f. Body breaks up into several fragments and each fragment starts to live as a new individual. This is fragmentation type of reproduction.
- g. Pollen grains are formed by meiotic division in locules of anthers.

3. Complete the paragraph with the help of words given in the bracket.

(Luteinizing hormone, endometrium of uterus, follicle stimulating hormone, estrogen, progesterone, corpus luteum)

Growth of follicles present in the ovary occurs under the effect of follicle stimulating hormone. This follicle secretes estrogen. Endometrium of uterus grows / regenerates under the effect of estrogen. Under the effect of Luteinizing hormone, fully grown-up follicle bursts, ovulation occurs and corpus luteum is formed from remaining part of follicle. It secretes progesterone and estrogen. Under the effect of these hormones, glands of endometrium of uterus are activated and it becomes ready for implantation.

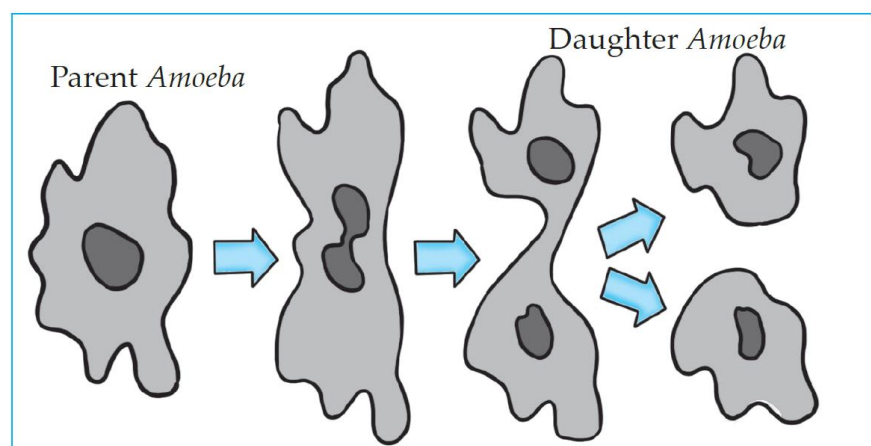
4. Answer the following questions in short.

a. Explain with examples types of asexual reproduction in unicellular organism.

- i. Process of formation of new organism by an organism of same species without involvement of gametes is called as asexual reproduction.
- ii. Types of asexual reproduction in unicellular organisms:

a. Binary fission:

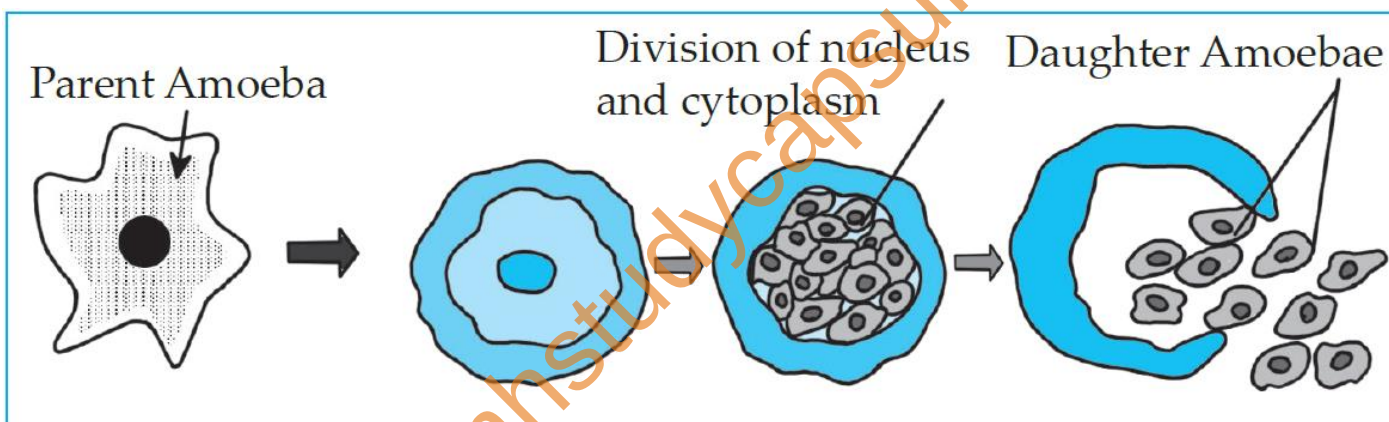
- i. In this process, the parent cell divides to form two similar daughter cells.
- ii. Prokaryotes (Bacteria), Protists (Amoeba, Paramecium, Euglena, etc.) and eukaryotic cell-organelle like mitochondria and chloroplasts perform asexual reproduction by binary fission.
- iii. Binary fission occurs either by mitosis or amitosis.
- iv. Axis of fission / division is different in different protists.
- v. Amoeba divides in any plane due to lack of specific shape; hence it is called as 'simple binary fission'.
- vi. Paramecium divides by 'transverse binary fission' whereas Euglena by 'longitudinal binary fission'.
- vii. Binary fission is usually performed by living organisms during favourable conditions i.e. availability of abundant food material.



To be drawn
in exam.

b. Multiple fission:

- i. Asexual reproduction by multiple fission is performed by Amoeba.
- ii. Amoeba stops the formation of pseudopodia whenever there is lack of food.
- iii. It becomes rounded and forms protective covering around plasma membrane. Such encysted Amoeba is called as cyst.
- iv. Many nuclei are formed by repeated nuclear division in the cyst.
- v. It is followed by cytoplasmic division and thus many amoebulae are formed.
- vi. They remain encysted till there are adverse conditions. Cyst breaks open on arrival of favourable conditions and many amoebulae are released.

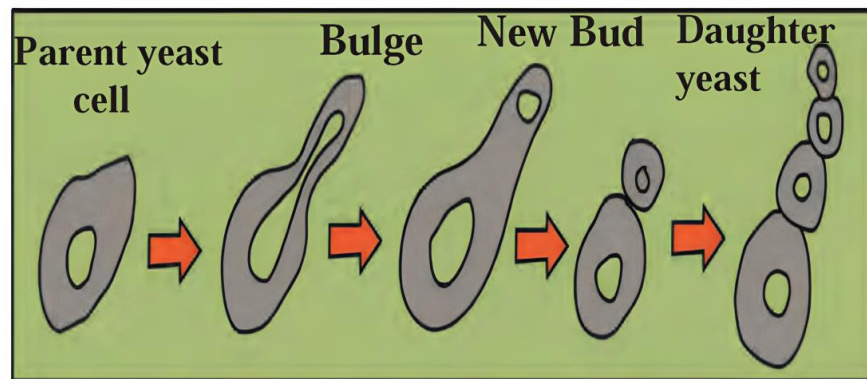


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in exam.

c. Budding:

- i. Asexual reproduction occurs by budding in yeast – a unicellular fungus.
- ii. Yeast cell produces two daughter nuclei by mitotic division. This yeast cell is called parent cell.
- iii. A small bulge appears on the surface of parent cell. This bulge is actually a bud.
- iv. One of the two daughter nuclei enters this bud.

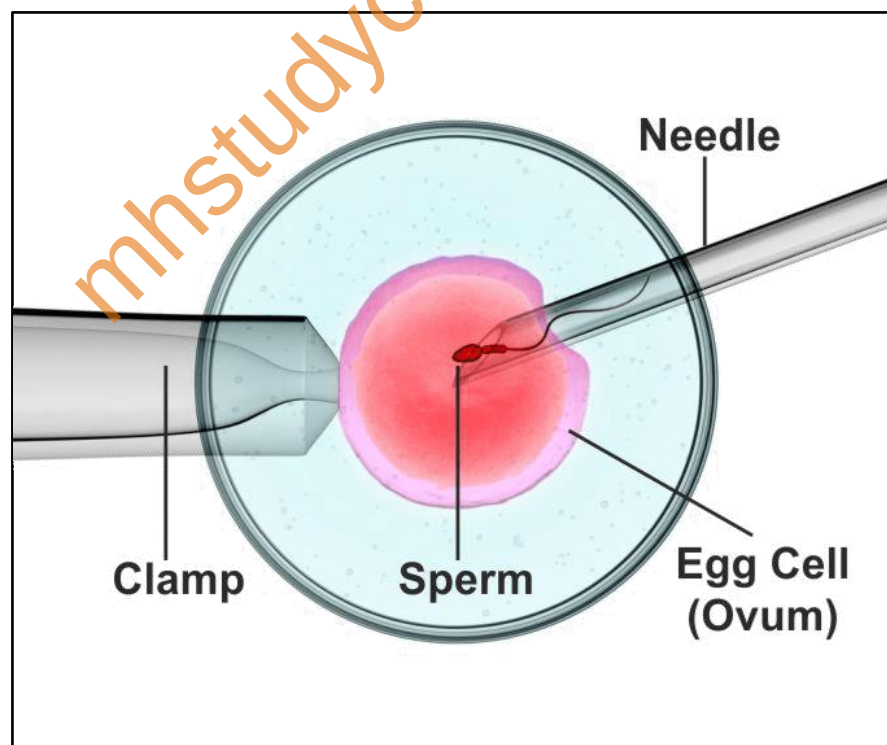
- v. After sufficient growth, bud separates from the parent cell and starts to live independently as a daughter yeast cell.



To be drawn
in exam.

b. Explain the concept of IVF.

- In IVF technique, fertilization is brought about in the test-tube and the embryo formed is implanted in uterus of woman at appropriate time.
- IVF technique is used for having the child in case of those childless couples who have problems like less sperm count, obstacles in oviduct, etc.

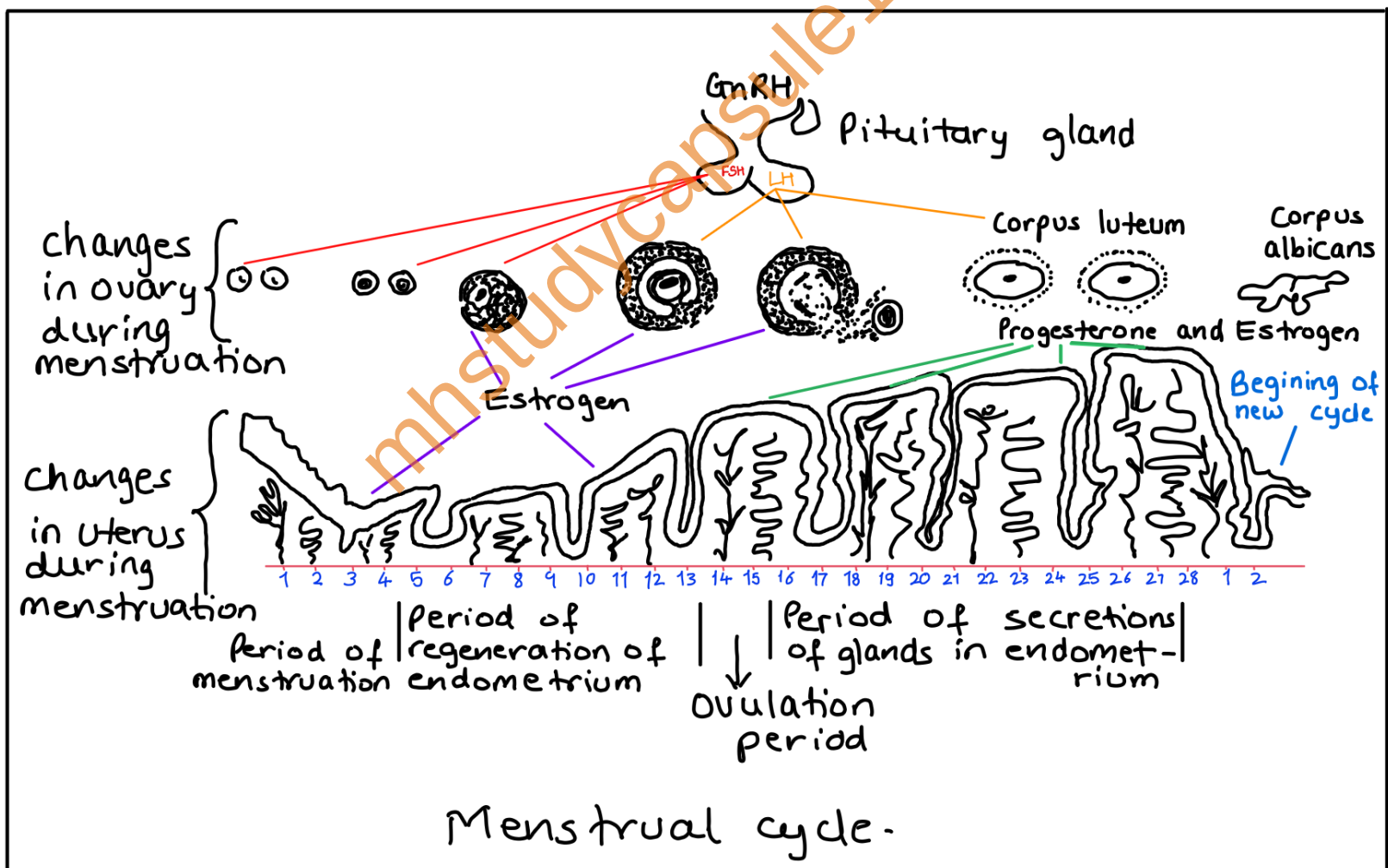


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reference.
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drawn.

c. Which precautions will you follow to maintain the reproductive health?

- i. A person's state of being physical, mental and social strength is called as health.
- ii. Bleeding occurs during menstrual cycle. Due to this, private organs need to be cleaned from time-to-time otherwise problem regarding reproduction health may arise.
- iii. It is essential to maintain the cleanliness of genitals of both males and females.
- iv. Also, one should avoid unsafe sexual contact.

d. What is menstrual cycle? Describe it in brief.



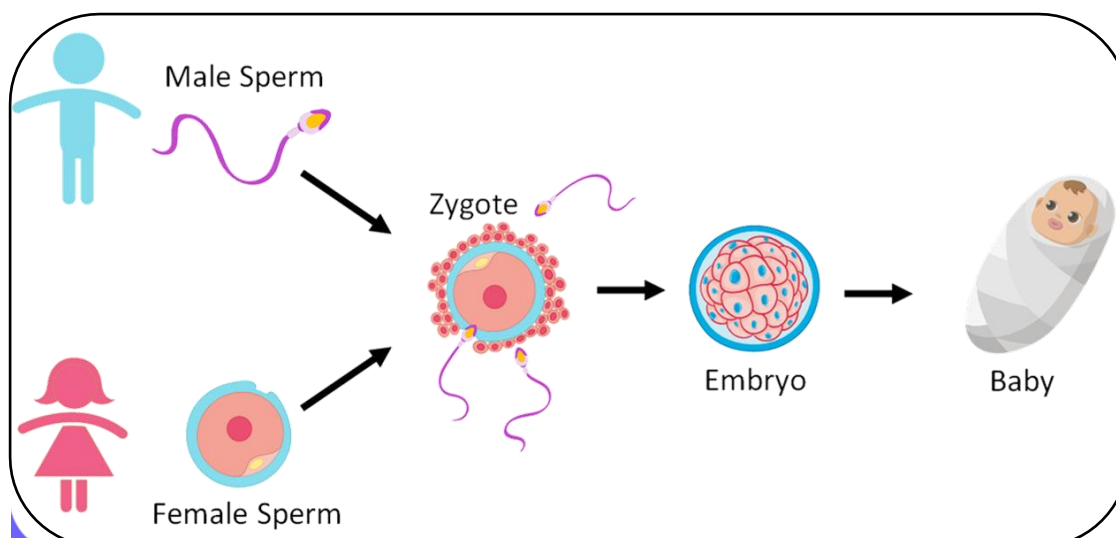
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- i. Female reproductive system undergoes some changes at puberty and those changes repeat at the interval of every 28 - 30 days. These repetitive changes are called as menstrual cycle.
- ii. Menstrual cycle is a natural process controlled by four hormones. Follicle stimulating hormone (FSH), Luteinizing hormone (LH), estrogen and progesterone.
- iii. One of the several follicles in the ovary starts to develop along with the oocyte present in it, under the effect of follicle stimulating hormone. This developing follicle secretes estrogen.
- iv. Endometrium of the uterus starts to develop or regenerate under the effect of estrogen.
- v. Meanwhile developing follicle completes its development.
- vi. It bursts under the effect of luteinizing hormone and oocyte is released. This is called as ovulation.
- vii. Remaining tissues of the burst follicle form the corpus luteum. Corpus luteum starts to secrete progesterone.
- viii. Endometrial glands secrete their secretion under the effect of progesterone. Such endometrium is ready for implantation of embryo.
- ix. If oocyte is not fertilized within 24 hours, corpus luteum becomes inactive and transforms into corpus albicans.
- x. Endometrium starts to degenerate in absence of estrogen and progesterone.

- xi. Tissues of degenerating endometrium and unfertilized ovum are discarded out through vagina, accompanied by bleeding.
- xii. Bleeding continues approximately for 5 days. This is called menstruation.

5. In case of sexual reproduction, new-born show similarities about characters. Explain this statement with suitable examples.

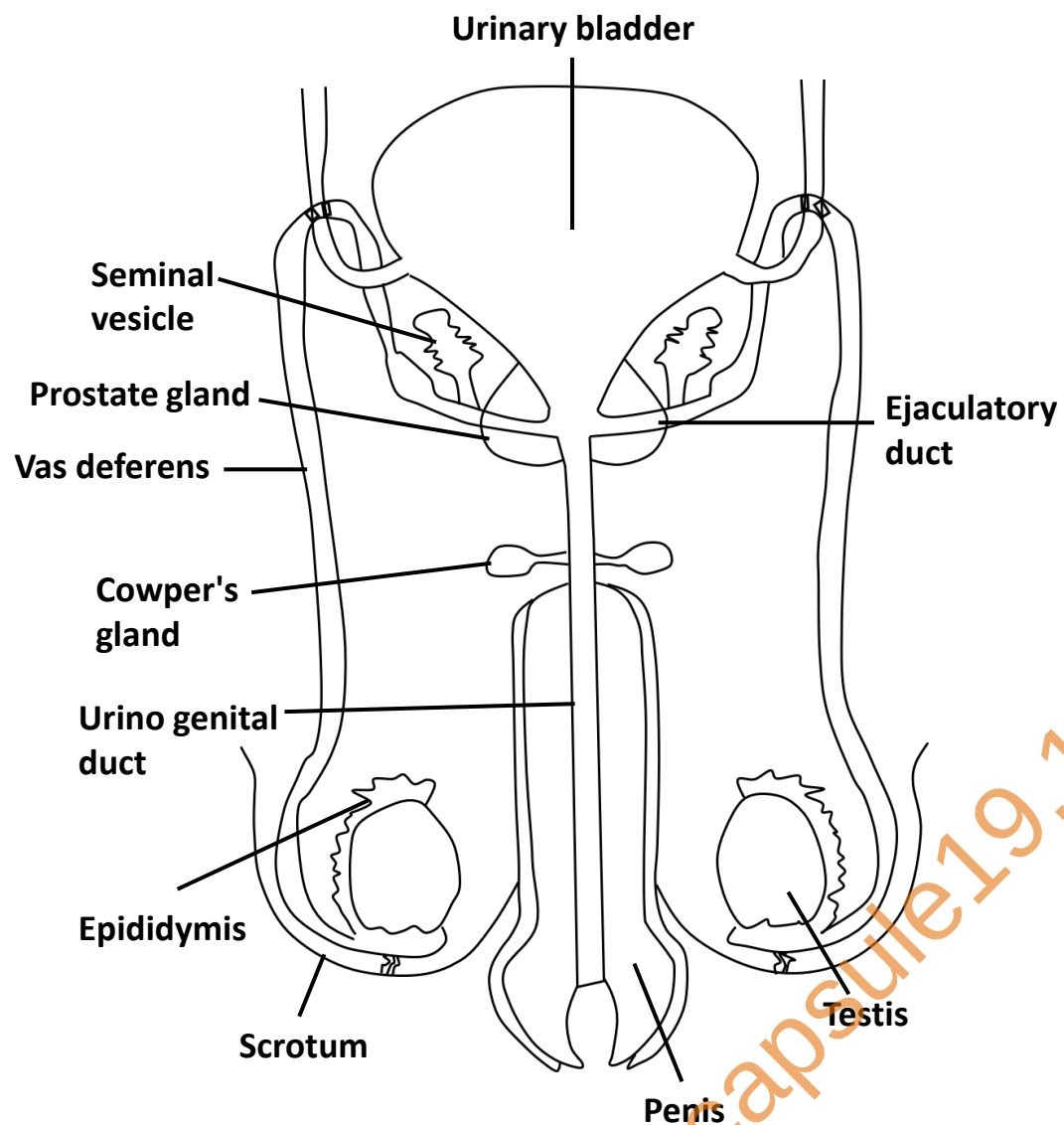
- i. Two parents i.e., male and female are involved in sexual type of reproduction.
- ii. The zygote formed contains a fusion of both the sperm nucleus and the egg nucleus.
- iii. Due to this, the new individual has characteristics that are similar to both the parents while some characteristics are different from both the parents.
- iv. Thus, the new individual formed from sexual reproduction inherits the characteristics of both the parents.
- v. If one parent has brown eyes and the other has blue eyes, their offspring might inherit a combination of these genes, resulting in a different shade of eye color, such as hazel or green.



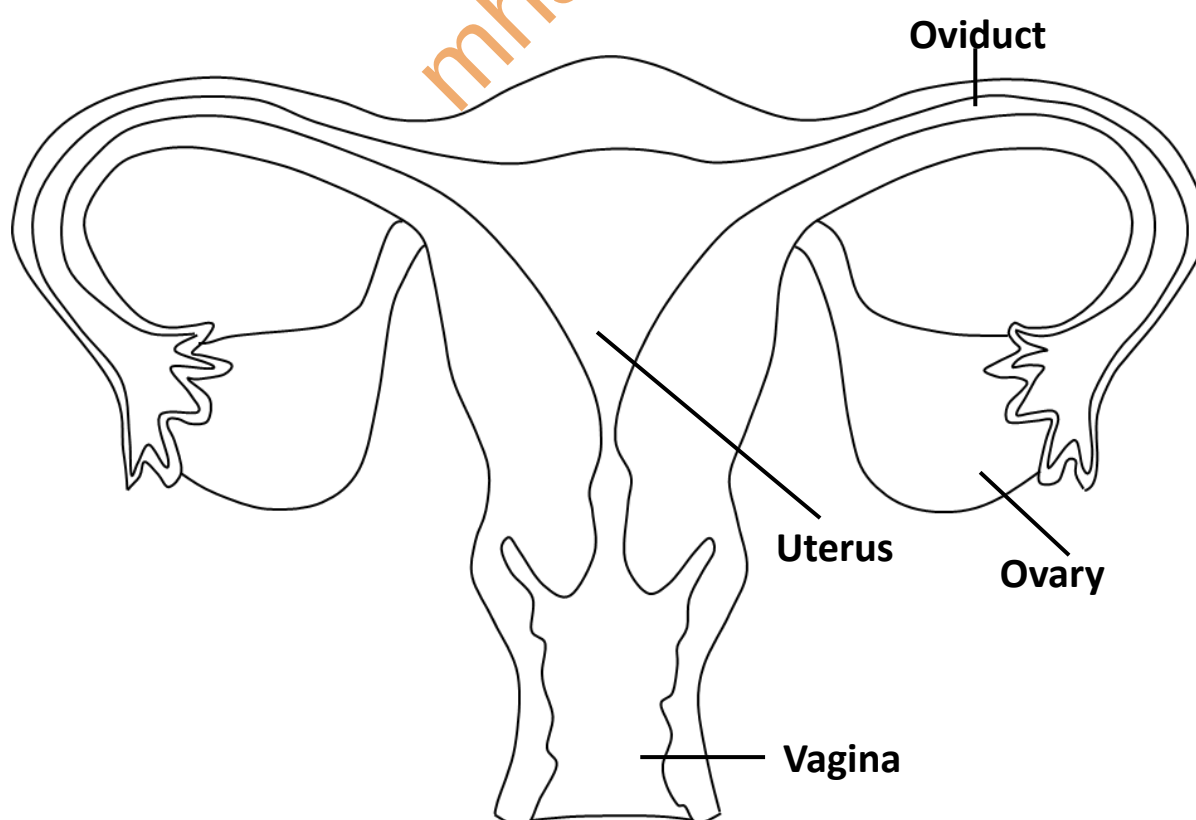
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6. Sketch the labelled diagrams.

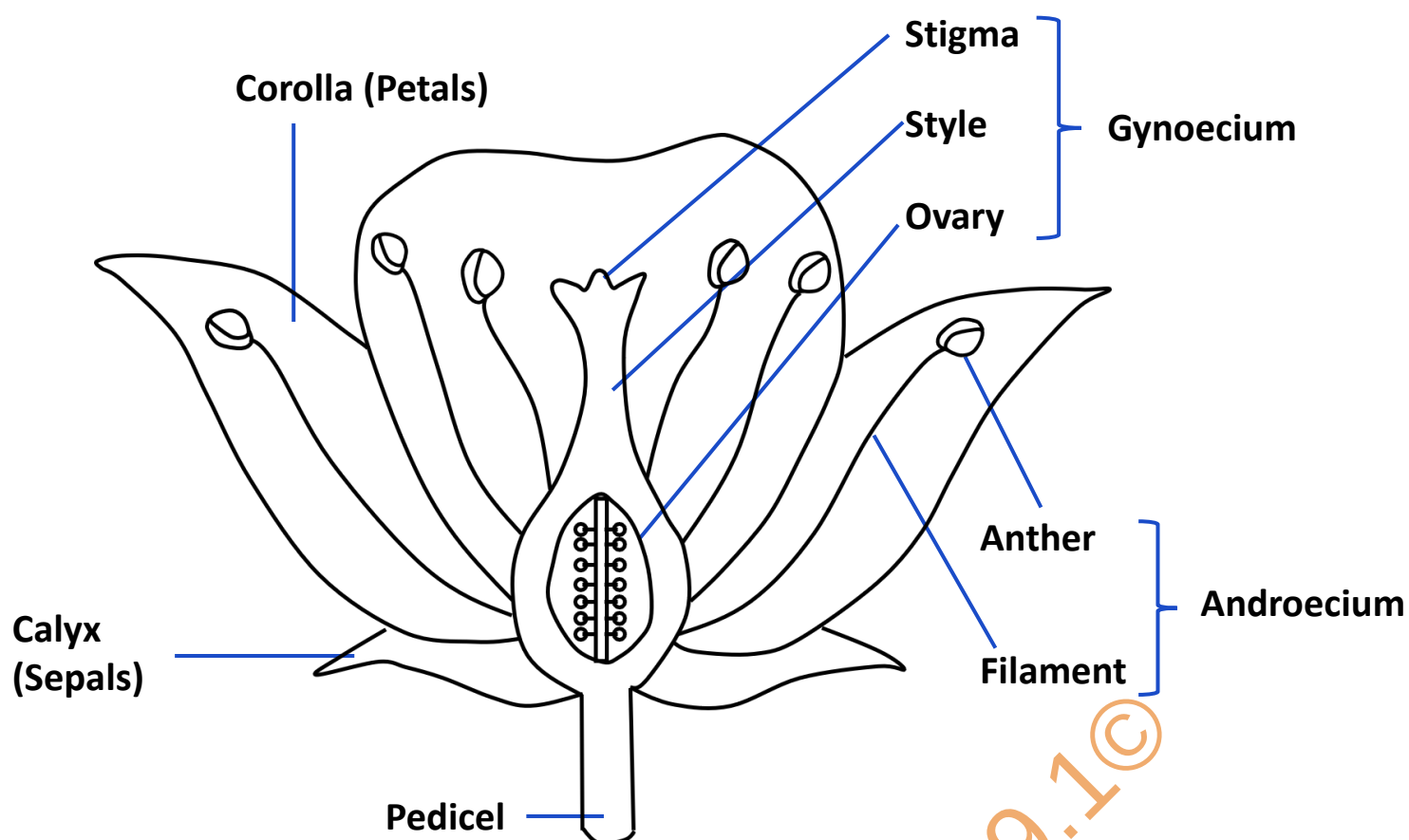
a. Human male reproductive system.



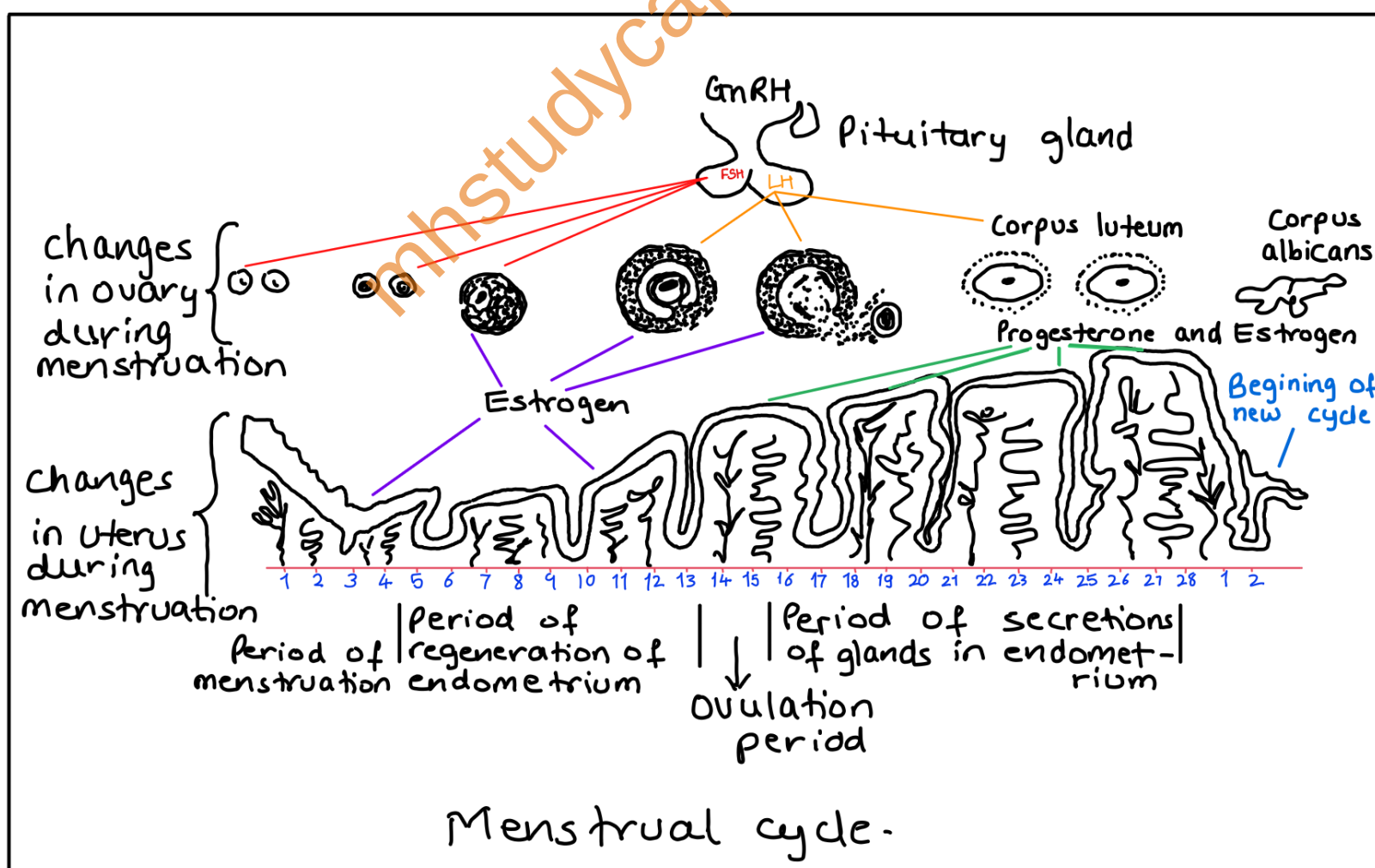
b. Human female reproductive system.



c. Flower with its sexual reproductive organs.



d. Menstrual Cycle



7. Give the names.

a. Hormones related with male reproductive system.

FSH (Follicle stimulating hormone), LH (Luteinizing hormone), testosterone.

b. Hormones secreted by ovary of female reproductive system.

Estrogen and progesterone

c. Types of twins.

Monozygotic twins and dizygotic twins.

d. Any two sexual diseases.

Syphilis and gonorrhoea

e. Methods of family planning.

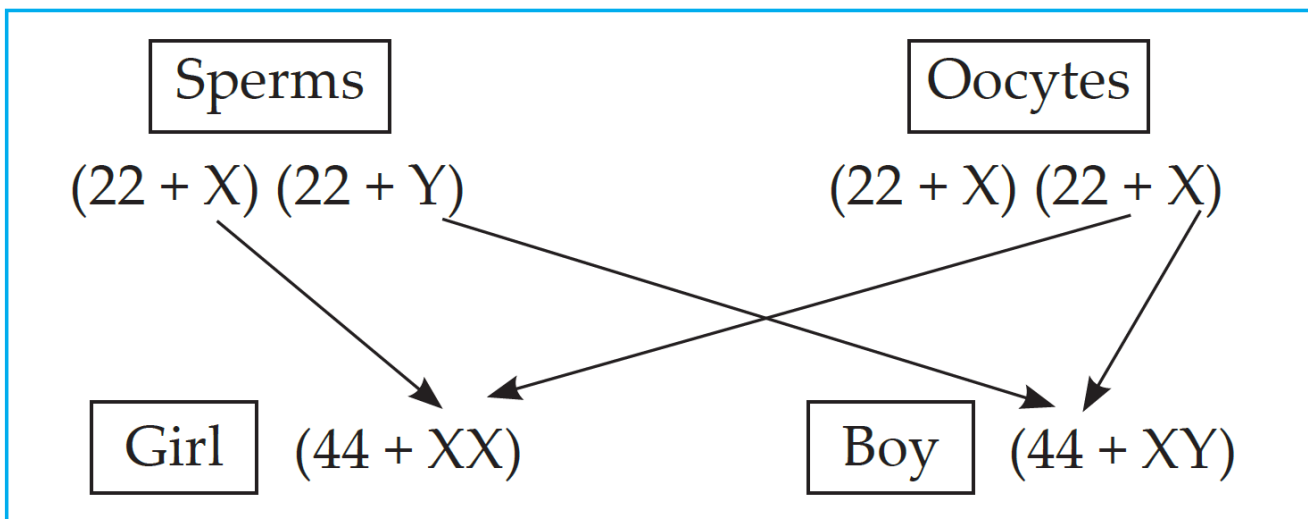
Intra uterine device, use of condoms

8. Gender of child is determined by the male partner of couple.

Explain with reasons whether this statement is true or false.

- i. After meiosis, two types of sperms are produced as $(22 + X)$ or $(22 + Y)$ whereas oocytes are produced of only one type as $(22 + X)$.
- ii. During zygote formation, man contributes either X or Y chromosome to the next generation, but woman contributes only X-sex chromosome to the next generation.
- iii. At the time of fertilization, if X- chromosomes comes from male, the child will be a girl and if Y-chromosome comes then the child will be a boy.

- iv. Thus, gender of child is determined by the male partner of the couple.
- v. So, the above statement is true.



To be drawn
in exam.

9. Explain asexual reproduction in plants.

- i. Reproduction in plants with the help of vegetative parts like root, stem, leaf and bud is called as vegetative reproduction.
- ii. Vegetative propagation in potatoes is performed with the help of eyes present on tuber whereas in *Bryophyllum*, it is performed with the help of buds present on leaf margin.
- iii. In case of plants like sugarcane and grasses, vegetative propagation occurs with the help of buds present on nodes.
- iv. Plants like carrot, radish and sweet potato perform vegetative propagation with the help of roots.
- v. Plants like *Spirogyra* reproduce asexually by fragmentation.
- vi. When there is plenty of water and nutrients available to *Spirogyra*, its filaments grow up very fast and break up into many small fragments. Each fragment starts to live independently as a new spirogyra fibre.

10. Modern techniques like surrogate mother, sperm bank and IVF technique will help the human beings. Justify this statement.

Nowadays, with the help of advanced medical techniques like IVF, Surrogacy, Sperm bank the childless couples can have a child.

a. In Vitro Fertilization (IVF)

- i. In this technique, fertilization is brought about in the test-tube and the embryo formed is implanted in uterus of woman at appropriate time.
- ii. IVF technique is used for having child in case of childless couples who have problems like less sperm count, obstacles in oviduct, etc.

b. Surrogacy

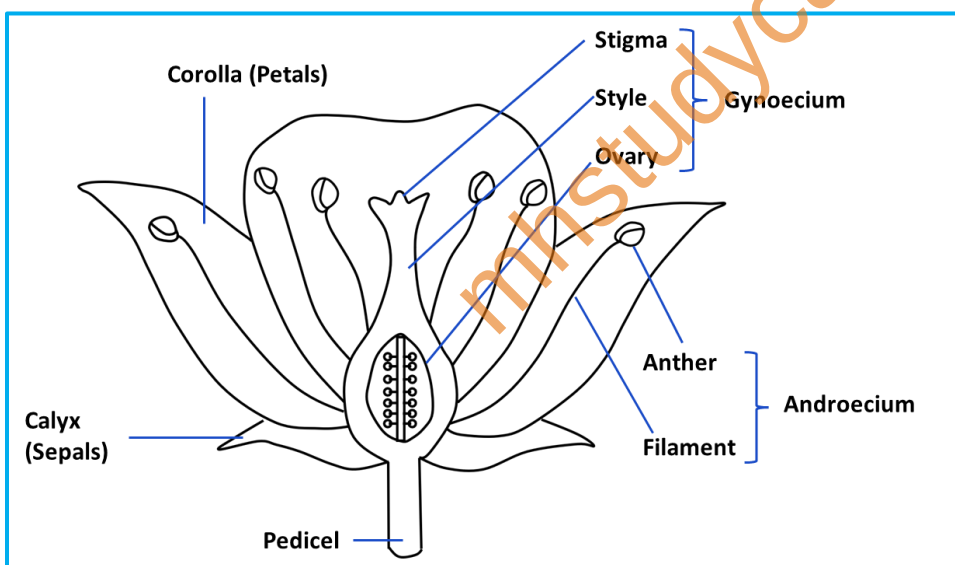
- i. Some women have problems in implantation of embryo in uterus. Such women can take the help of the modern remedial technique called as surrogacy.
- ii. In this technique, oocyte is collected from the ovary of the woman having problem in implantation in uterus.
- iii. That oocyte is fertilized in test-tube with the help of sperms collected from her husband.
- iv. The embryo formed from such fertilization is implanted in the uterus of some other woman having normal uterus.
- v. Such a woman, in whose uterus the embryo is implanted, is called as surrogate mother.

c. Sperm Bank/ Semen Bank

- i. There are various problems in sperm production in case of many men.
- ii. So as to have children in case of such couples, new concept of sperm bank has been introduced. This concept is similar to blood bank.
- iii. Semen ejaculated by the desired men is collected after their thorough physical and medical check-up and stored in the sperm bank.
- iv. As per the wish of needful couple, oocyte of woman of the concerned couple is fertilized by IVF technique using the semen from sperm bank.
- v. Resultant embryo is implanted in the uterus of same woman.

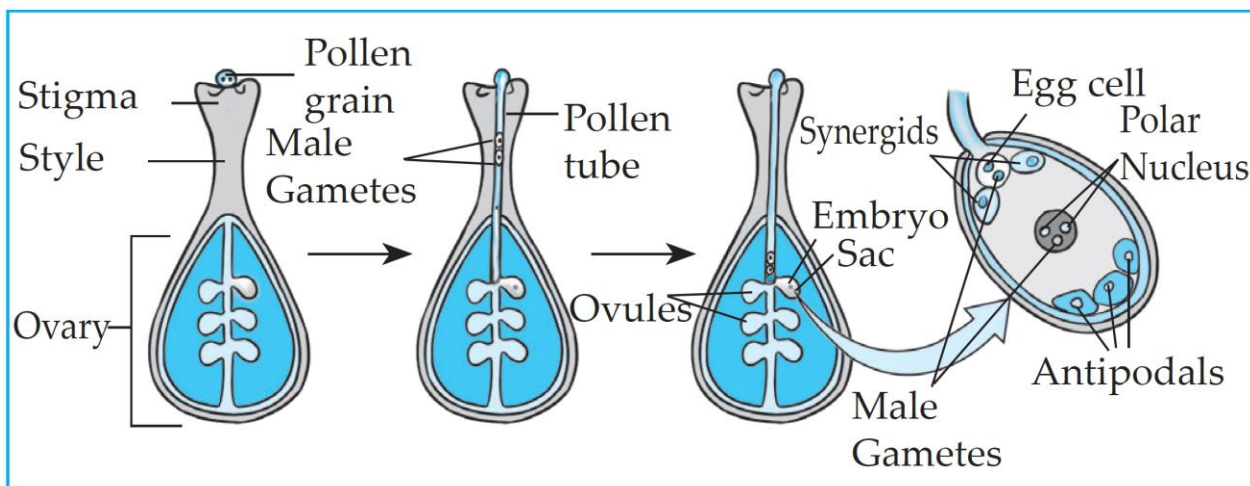
Thus, modern techniques like IVF, surrogacy and sperm bank help human beings.

11. Explain sexual reproduction in plants.



Draw both diagrams for this question.

Only draw this if double fertilisation is asked.



- i. A flower is the site of sexual reproduction in plants.
- ii. It consists of four floral whorls calyx, corolla, androecium and gynoecium; arranged in sequence from outside to inside.
- iii. Androecium and gynoecium are called as essential whorls because they perform the function of reproduction
- iv. Calyx and corolla are called as accessory whorls because they are responsible for protection of inner whorls.
- v. Androecium is male whorl, and its members are called stamens. Gynoecium is female whorl, and its members are called as carpels.
- vi. Ovary present at the basal end of carpel contains one or many ovules. A hollow style comes up from the ovary. Stigma is present at the tip of style.
- vii. Ovary contains one or many ovules. Embryo sac is formed in each ovule by meiosis and each embryo sac consists of a haploid egg cell and two haploid polar nuclei.
- viii. Pollen grains are transferred from anther to the stigma. This is called as pollination.
- ix. Stigma becomes sticky during pollination. Pollens germinate when they fall upon such sticky stigma i.e., a long pollen tube and two male gametes are formed..
- x. The pollen tube carries male gametes. Pollen tube reaches the embryo sac via style.

- xi. Tip of the pollen tube bursts, and two male gametes are released in embryo sac. One male gamete unites with egg cell to form zygote. This process is called fertilization.
- xii. The second male gamete unites with two polar nuclei and endosperm is formed.
- xiii. As two male nuclei participate in this process, it is called as double fertilization.
- xiv. After fertilization ovule develops into seed and ovary into fruit.
- xv. Seeds fall upon the ground when fruits break up and they germinate in the soil under favourable conditions.
- xvi. Zygote develops at the cost of food stored in endosperm of seed and thus a new plantlet is formed. This is called as seed germination.

Start answer from point (viii) in case if double fertilization is asked.

INTEXT QUESTIONS

1. Which are the important life processes in living organisms?

Control and co-ordination, respiration, growth, excretion, reproduction, nutrition and excretion are some of the important life processes in living organisms.

2. Which life processes are essential for production of energy required by body?

Nutrition and respiration are the processes essential for production of energy.

3. Which are main types of cell-division? What are the differences?

- i. The two main types of cell division are 'mitosis' and 'meiosis'.
- ii. Mitosis occurs in somatic cells and stem cells of the body, whereas meiosis occurs in germ cells.
- iii. In mitosis, the chromosome number remains the same and two daughter cells are obtained from one cell.
- iv. In meiosis, chromosome number is reduced to half and four daughter cells are obtained from one cell.

4. What is the role of chromosomes in cell-division?

In cell division, chromosomes carry genetic information from parent cell to daughter cells.

5. What do we mean by maintenance of species?

- i. The process of reproduction helps to increase the number of organisms of a species by which the species can prevent itself from being extinct.
- ii. This is called as maintenance of species.

6. Whether the new organism is genetically exactly similar to earlier one that has produced it?

Offsprings are genetically similar to parents if they reproduce asexually and offsprings are genetically different if they reproduce sexually.

7. Who determines whether the two organism of a species will be exactly similar or not?

Method of reproduction will determine whether the two organisms of species will be exactly similar or not.

8. What is the relationship between the cell division and formation of new organism of same species by earlier existing organism?

- i. In case of asexual reproduction, the parent cell divides to form two similar daughter cells. This occurs by the type of cell division called mitosis.
- ii. In case of sexual reproduction, gametes are formed by the type of cell division called meiosis. These gametes fuse to form a new organism.

9. Does the parent cell exist after asexual reproduction- fission?

- i. In asexual reproduction - Fission, the mother cell divides into daughter cells and each daughter cell begins its life like a new individual.
- ii. The parent cell thus does not exist after asexual reproduction - fission.

10. What would have been happened if the male and female gametes had been diploid?

- i. If the male and female gamete were diploid, then the number of chromosomes in each generation would have continued to double and this would result in an increasing number of chromosomes.
- ii. The number of chromosomes in a particular species should remain a constant generation after generation.

11. What would have been happened if any of the cells in nature had not been divided by meiosis?

- i. Meiosis is necessary for sexual reproduction and for increasing genetic diversity.
- ii. If any of the cells in nature had not been divided by meiosis, then all sexually reproducing animals would have no offspring.
- iii. Only asexually reproducing animal could have survived on his Earth. This would lead to the extinction of the advanced diploid species.

12. Which different hormones control the functions of human reproductive system through chemical coordination?

- i. FSH (follicle stimulating hormone) LH (luteinizing hormone)
- ii. Male hormone - testosterone
- iii. Female hormones - estrogen and progesterone are the different hormones that control the functions of human reproductive system through chemical co-ordination.

13. Which hormones are responsible for changes in human body occurring during onset of sexual maturity?

The hormones responsible for changes in human body during onset of sexual maturity are testosterone in males and estrogen and progesterone in females.

14. Why has the Government of India enacted the law to fix the minimum age of marriage as 18 in girls and 21 in boys?

- i. The Government of India enacted the law to fix the minimum age of marriage as 18 for girls and 21 for boys primarily to promote gender equality, protect the well-being of young individuals, and address social and health concerns.
- ii. Early pregnancies in young girls can pose serious health risks due to their physical immaturity. By setting a minimum age for marriage, the government aims to reduce maternal and infant mortality rates, as well as improve the overall health and well-being of young brides.

- iii. Early marriages, especially of girls, often lead to negative outcomes such as reduced educational and economic opportunities, increased health risks for young mothers and infants, and perpetuation of poverty. By setting a minimum age, the government seeks to discourage early marriages and promote education and personal development.

15. Which hormone is released from pituitary of mother once the foetal development is completed?

Oxytocin is the hormone released from pituitary of the mother, once fetal development is completed.

16. Under the effect of that hormone, which organ of the female reproductive system starts to contract and thereby birth process (Parturition) is facilitated?

Uterus starts to contract under the effect of oxytocin hormone and facilitates parturition

Previous Year Questions

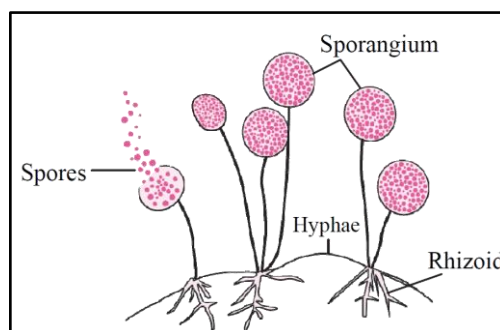
1. Answer the following questions:

- a. "Gender of child is determined by the male partner of couple."

Draw a diagram explaining the above statement.

- b. Prepare a slogan for campaign against female foeticide.

- c. In the following figure, explain how new fungal colonies of mucor are formed:



- d. Identify and state the type of reproduction represented in the above figure.

a. Exercise Question 8

- b. "Embrace Equality, Protect Her Potential: Say No to Female Foeticide!"
- c. Mucor reproduced asexually by spore formation. It has filamentous body that possess sporangia. Once the spores are formed, sporangia burst, and spores are released. Spores germinate in moist and warm place and new fungal colony is formed.
- d. The above figure shows asexual mode of reproduction by spore formation.

2. Despite of various diagnostic tests, a couple could not have a child. In this situation, which remedies will you suggest?

Despite of various diagnostic tests, a couple could not have a child, then they have to resort to anyone of the following measure :

- i. **IVF**: IVF or In Vitro Fertilization is the technique used when a couple has a problem of either low sperm count or obstacles in oviduct. Fertilization is brought about in test-tube, and the embryo is formed implanted in uterus of woman at appropriate time.
- ii. **Surrogacy**: Women who have problems in implantation of embryo in uterus, can take the help of the modern remedial technique called as surrogacy. Embryo formed from IVF is implanted in the uterus of some other woman having normal uterus. Such a woman is called as surrogate mother.
- iii. **Sperm bank or semen bank**: For men having problems in sperm production, new concept of sperm bank has been introduced. As per the wish of needful couple, oocyte of woman of the concerned couple is fertilized by IVF technique using the semen from sperm bank. Resultant embryo is implanted in the uterus of same woman.



3. Answer the following questions:

- a. In our country, there seems to be lack of awareness regarding reproductive health. Why?**
- b. Write the symptoms of disease gonorrhea.**
- c. What precautions will you take maintain reproductive health?**

- a. In our country, there seems to be lack of awareness regarding reproductive health due to various reasons like social customs, traditions, illiteracy, shyness, etc.
- b. Painful and burning sensation during urination, oozing of pus through penis and vagina, inflammation of urinary tract, anus, throat, eyes, etc. are symptoms of gonorrhoea.
- c. Precautions which should be taken in order to maintain good reproductive health are:
 - i. Eat a balanced and nutritious diet
 - ii. Cleanliness of genitals and personal hygiene
 - iii. Adequate physical exercise
 - iv. Avoid taking any drugs

4. Indians should follow family planning for controlling the population.

- i. Excessive growth of population within short duration is called as population explosion.
- ii. Because of population explosion in India, we have to face various problems like unemployment, decreasing per capita income and increasing loan, stress on natural resources, etc.

- iii. There is only one solution for all such problems, and it is population control.
- iv. Family planning is essential for this.

5. Explain the following concepts in short: Exercise Question 10

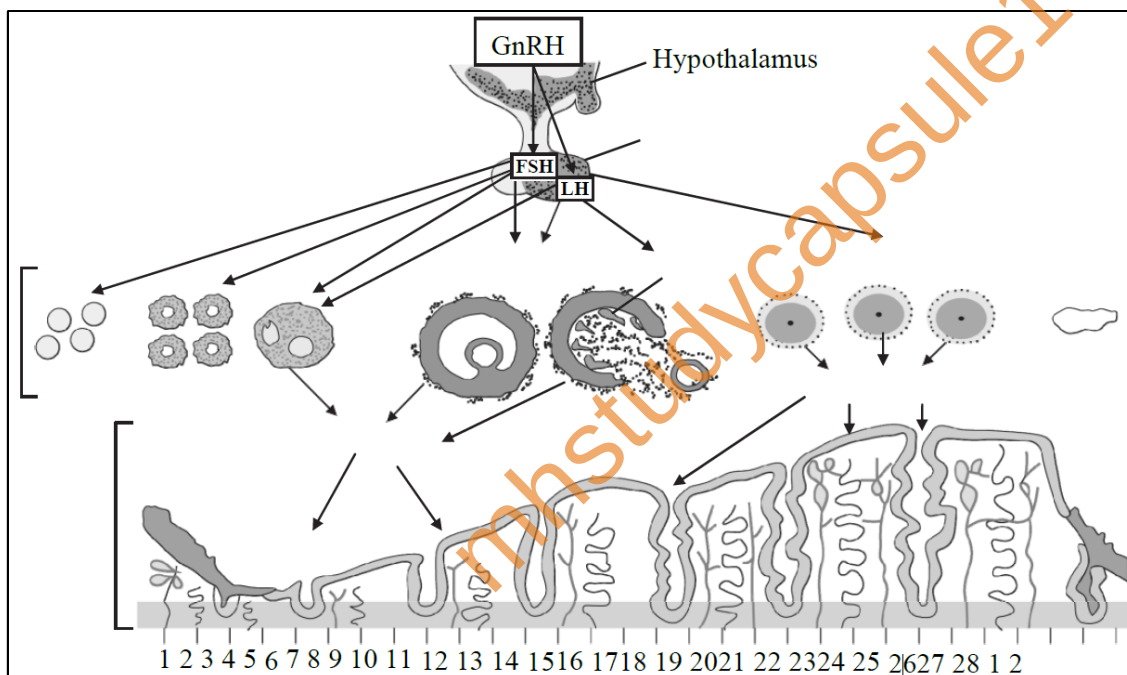
- a. Surrogacy
- b. In Vitro Fertilization (IVF)
- c. Sperm Bank

6. Explain the term:

Modern technology of reproduction IVF.

Explain IVF from Exercise Question 10

7. Observe the following figure and answer the questions:



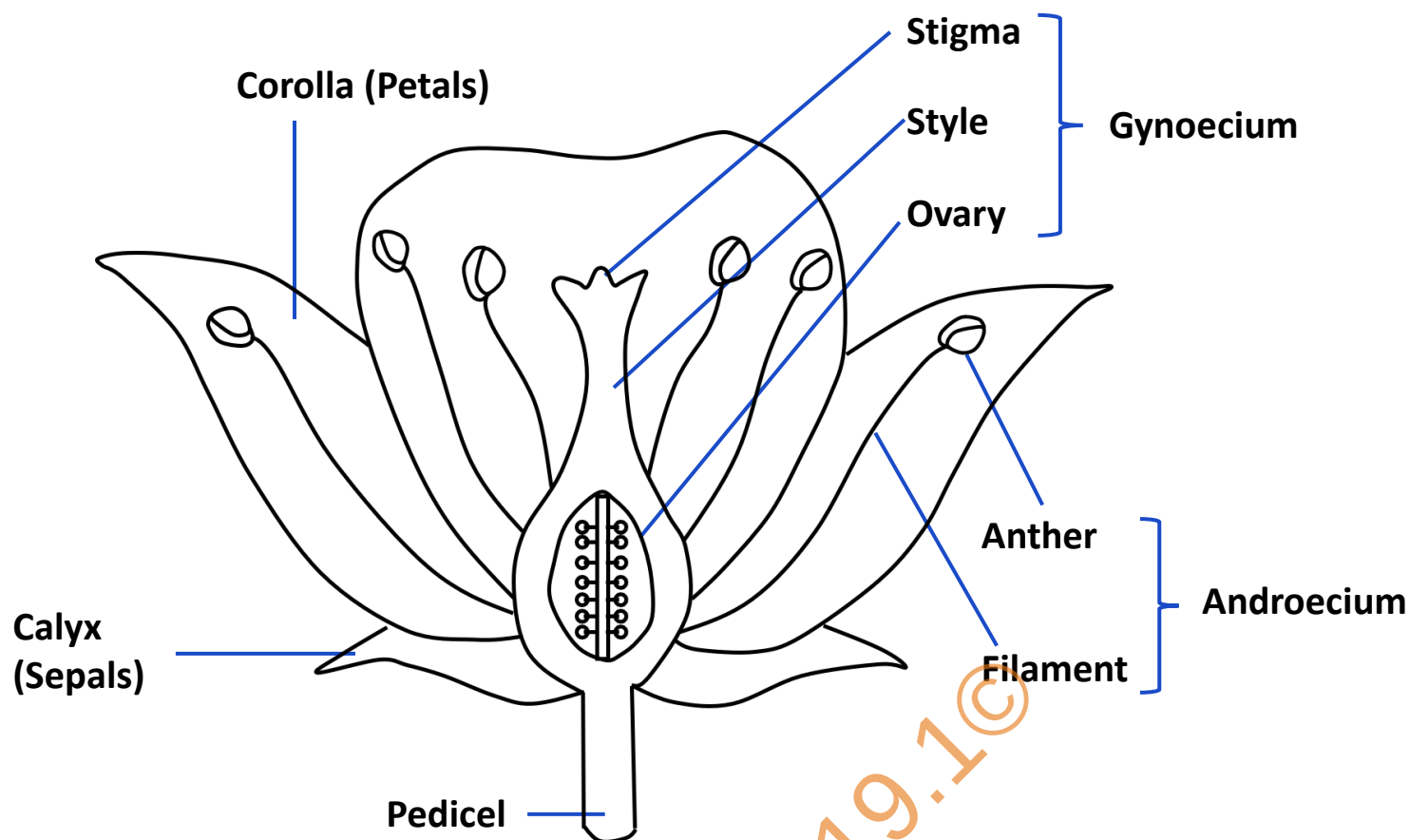
- a. Identify the process in above figure and after how many days it will repeat again?

The above figure shows Menstrual Cycle. It repeats every 28-30 days.

- b. Write the names of hormones involved in this process.

The hormones involved in this process are FSH (follicle stimulating hormone), LH (luteinizing hormone), estrogen and progesterone.

8. With the help of a neat labelled diagram write any three parts of inside of flower.



- i. Flower is structural unit of sexual reproduction in plants. It consists of four floral whorls as calyx, corolla, androecium and gynoecium; arranged in sequence from outside to inside.
- ii. Androecium and gynoecium are called 'essential whorls' because they perform the function of reproduction whereas calyx and corolla are called as 'accessory whorls' because they are responsible for protection of inner whorls.
- iii. Accessory Whorls: Members of calyx are called as 'sepals' and they are green coloured. Members of corolla are called as 'petals' and they are variously colored.

- iv. Essential Whorl – Androecium: It is male whorl, and its members are called as stamens. Stamens consist of anther and filament. Pollen grains are formed by meiotic division inside locules of anthers.
- v. Essential Whorl – Gynoecium: It is female whorl, and its members are called as carpels. These may be separate or united. Ovary is present at the basal end of each carpel. A hollow 'style' comes up from the ovary. Stigma is present at the tip of style. Ovary contains one or many ovules. Embryo sac is formed in each ovule by meiosis. Each embryo sac consists of a haploid egg cell and two haploid polar nuclei.

9. Distinguish between sexual reproduction and asexual reproduction.
(two points) – Exercise Question 1

10. Answer the following:

- a. **Sketch and label the essential and accessory whorls of flower. – Q8**
- b. **What is pollination?**

Transfer of pollen grains from anther to stigma is called as pollination.

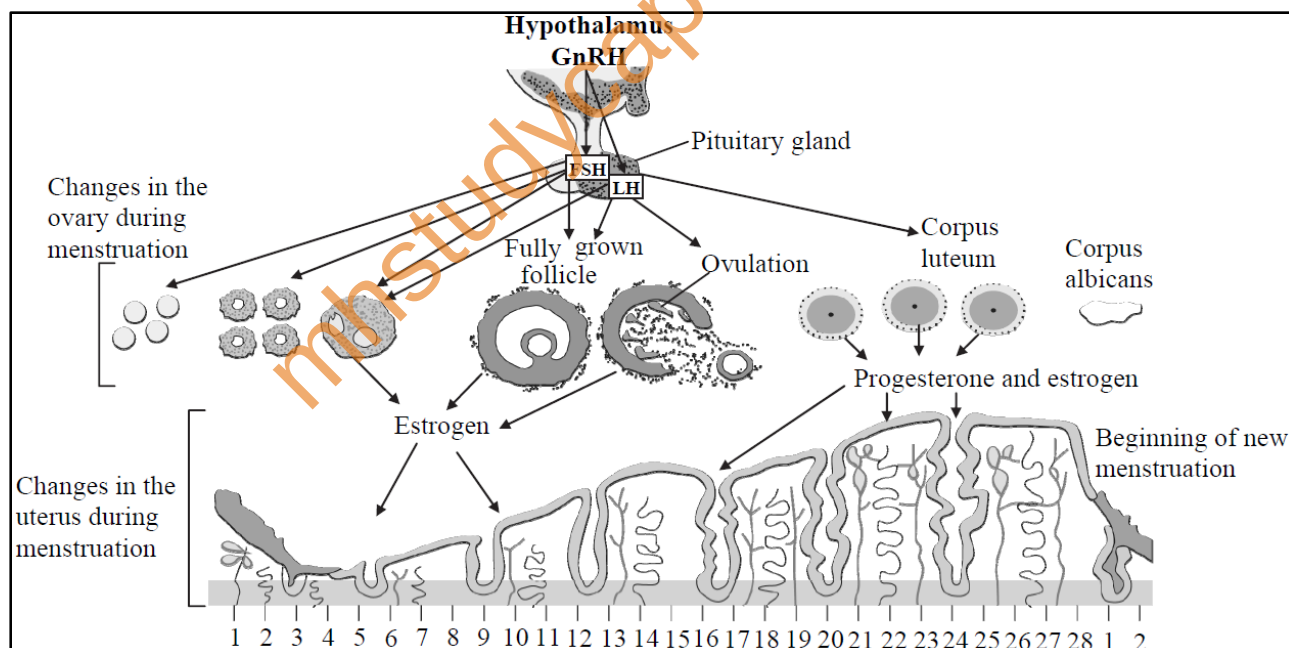
- c. **Give any two examples of agents of pollination.**

Wind, water

11. Flower is structural unit of sexual reproduction in plants.

- Flower consists of four floral whorls as calyx, corolla, androecium and gynoecium; arranged in sequence from outside to inside.
- Androecium and gynoecium are called 'essential whorls' because they perform the function of reproduction.
- Calyx and corolla are called as 'accessory whorls' because they are responsible for protection of inner whorls.
- Since the essential whorls - androecium and Gynoecium perform the function of sexual reproduction, flower is called as structural unit of sexual reproduction in plants.

12. Answer the following questions by observing the diagram of menstrual cycle:



a. What is the period of menstruation?

Period of menstruation is from about 1st – 5th day of the cycle.

b. On which day does ovulation occur during menstrual cycle?

Ovulation occurs around 14th or 15th day of the menstrual cycle.

c. In menstrual cycle which reproductive organs undergo changes?

Ovary and Uterus undergo changes in menstrual cycle.

d. Which period is said to be period of regeneration of endometrium?

In the menstrual cycle, days 5 to 14 are the period of regeneration of the endometrium.

e. Which period is said to be period of secretions of glands in endometrium?

The period of secretions of glands in the endometrium is 15th to 28th day of the cycle.

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